Numerical Methods

Chapter Overview

Solution 2

At GCSE you used iteration to obtain successively better approximations to solutions. We revisit this and introduce a more powerful method.

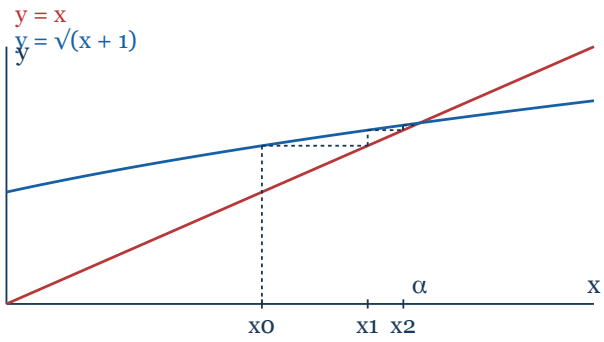
1. Locating roots

Understand what a root is and when a root must lie in an interval.

Show that $f(x) = x^3 - 4x^2 + 3x + 1$ has a root between $x = 1.4$ and $x = 1.5$.

2. Iteration

Rearrange $f(x) = 0$ as $x = g(x)$, then use $x_{n+1} = g(x_n)$.



3. Newton–Raphson

Follow tangents to approach a root, often faster:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Why Do We Need Numerical Methods?

Solution 3

Finding a root of $f(x)$ means finding an input whose output is zero.

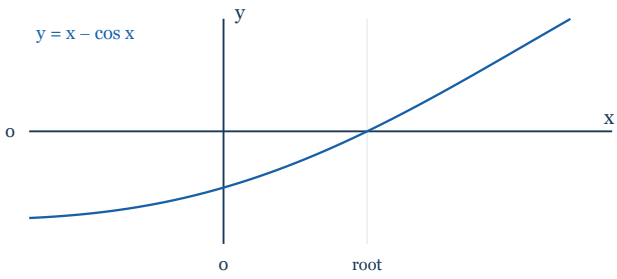
Solve $f(x) = 0$.

Some exact roots are complicated and difficult to calculate. For example, $x^3 + 2x^2 - 3x + 4 = 0$:

$$x = \frac{1}{3}\left(-2 - \frac{13}{\sqrt[3]{89 - 6\sqrt{159}}} - \sqrt[3]{89 - 6\sqrt{159}}\right).$$

Other equations, such as $x - \cos x = 0$, have no standard exact algebraic formula for their root.

Use a numerical approximation: the root is approximately 0.739085.

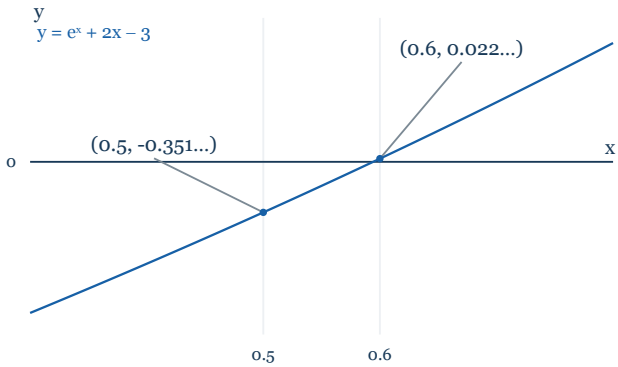


Numerical methods give progressively better approximations to solutions of $f(x) = 0$.

Proving a Solution Lies in an Interval

Solution 4

Show that $f(x) = e^x + 2x - 3$ has a root between $x = 0.5$ and $x = 0.6$.



If a continuous graph changes from negative to positive, its y-values must pass through zero.

Root interval

$f(0.5) = e^{0.5} + 1 - 3 = -0.351278\ldots < 0,$

$f(0.6) = e^{0.6} + 1.2 - 3 = 0.022118\ldots > 0.$

There is a change of sign, and f is continuous on $[0.5, 0.6]$. Hence a root lies between 0.5 and 0.6.

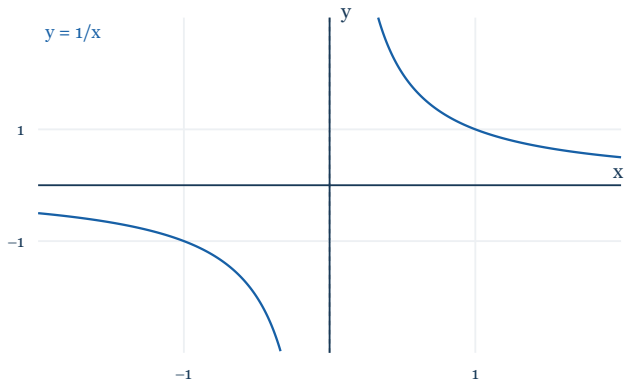
Exam tip

Calculate both endpoint values and explicitly state that there is a change of sign. Include continuity in the argument.

The Function Must Be Continuous

Solution 5

A student claims: "For $f(x) = 1/x$, $f(-1) = -1$ and $f(1) = 1$. There is a change of sign, so a root lies in $[-1, 1]$."



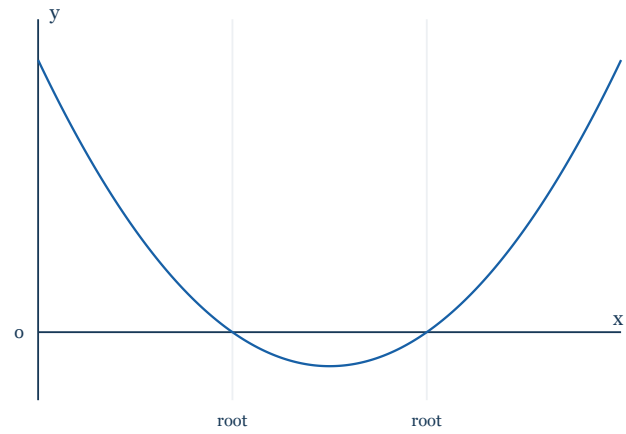
Why is this claim wrong?

The function is not continuous across this interval: it is undefined at $x = 0$, where there is a vertical asymptote. It never equals zero.

A sign change guarantees a root only when the function is continuous throughout the interval. Otherwise the graph can pass from negative to positive without crossing the axis.

No Sign Change Does Not Mean No Root

Solution 6



No sign change does **not** prove that an interval contains no roots.

The graph can cross the axis twice and return to the original sign. A graph can also touch the axis at a root without changing sign.

The sign-change test can therefore miss roots: it gives a sufficient condition when continuity holds, not a necessary condition.

Proving a Solution to a Given Accuracy

Solution 7

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$g(x) = e^{x-1} + x - 6$. Its root is α.

c. By choosing a suitable interval, show that $\alpha = 2.307$ correct to 3 decimal places.

What are the rounding boundaries?

Lower boundary

Lower boundary: 2.3065.

Upper boundary

Upper boundary: 2.3075.

A root strictly between these boundaries rounds to 2.307 to 3 decimal places.

c

$g(2.3065) = -0.000275221\ldots < 0,$

$g(2.3075) = 0.004419850\ldots > 0.$

There is a change of sign, and g is continuous. Thus

$2.3065 < \alpha < 2.3075.$

Therefore $\alpha = 2.307$ to 3 decimal places.

Example — Logarithm and Reciprocal

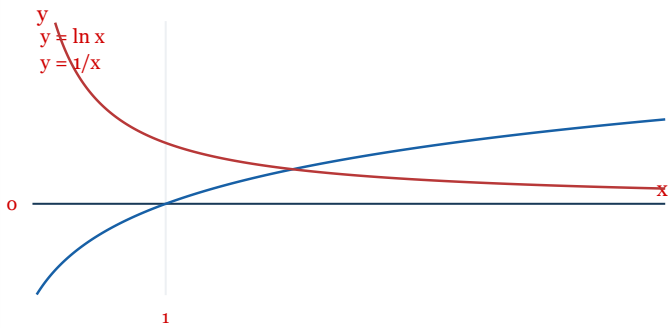
Solution 8

a. Sketch $y = \ln x$ and $y = 1/x$ on the same axes. Explain why $f(x) = \ln x - 1/x$ has only one root.

b. Show that the root lies in $1.7 < x < 1.8$.

c. Show that $\alpha = 1.763$ to 3 decimal places.

a sketch



a

Intersections satisfy $\ln x = 1/x$, hence $f(x) = 0$. The logarithm increases and the reciprocal decreases for $x > 0$; they meet once.

b

$f(1.7) = -0.057607\ldots < 0,$

$f(1.8) = 0.032231\ldots > 0.$

Continuity and a sign change give a root in $(1.7, 1.8)$.

c

$f(1.7625) = -0.000642630\ldots < 0,$

$f(1.7635) = 0.000246317\ldots > 0.$

Continuity and a sign change give $1.7625 < \alpha < 1.7635$, so $\alpha = 1.763$ to 3 d.p.

Exercise 10A — Pages 276–277 Solution 9

2. $f(x) = 3 + x^2 - x^3$.

a. Show that a root α lies in $[1.8, 1.9]$.

b. Use a sign change in a suitable interval to verify $\alpha = 1.864$ to 3 decimal places.

2a

$f(1.8) = 3 + 1.8^2 - 1.8^3$

$= 0.40800000\ldots,$

$f(1.9) = 3 + 1.9^2 - 1.9^3$

$= -0.24900000\ldots.$

The values have opposite signs, and the function is continuous on the interval. Hence a root lies between 1.8 and 1.9.

2b

Test the rounding boundaries:

$f(1.8635) = 3 + 1.8635^2 - 1.8635^3$

$= 0.00138205\ldots,$

$f(1.8645) = 3 + 1.8645^2 - 1.8645^3$

$= -0.00531344\ldots.$

There is a change of sign. Continuity gives $1.8635 < \alpha < 1.8645$, so $\alpha = 1.864$ to 3 decimal places.

Exercise 10A — Pages 276–277 Solution 10

3. $h(x) = \sqrt[3]{x} - \cos x - 1$. Use radians.

a. Show that a root α lies in $[1.4, 1.5]$.

b. Use a sign change in a suitable interval to verify $\alpha = 1.441$ to 3 decimal places.

3a

$h(1.4) = \sqrt[3]{1.4} - \cos 1.4 - 1$

$= -0.05127820\ldots,$

$h(1.5) = \sqrt[3]{1.5} - \cos 1.5 - 1$

$= 0.07397704\ldots.$

The values have opposite signs, and the function is continuous on the interval. Hence a root lies between 1.4 and 1.5.

3b

Test the rounding boundaries:

$h(1.4405) = \sqrt[3]{1.4405} - \cos 1.4405 - 1$

$= -0.00055404\ldots,$

$h(1.4415) = \sqrt[3]{1.4415} - \cos 1.4415 - 1$

$= 0.00069882\ldots.$

There is a change of sign. Continuity gives $1.4405 < \alpha < 1.4415$, so $\alpha = 1.441$ to 3 decimal places.

Exercise 10A — Pages 276–277 Solution 11

4. $f(x) = \sin x - \ln x$. Use radians. The domain is $x > 0$.

a. Show that a root α lies in $[2.2, 2.3]$.

b. Use a sign change in a suitable interval to verify $\alpha = 2.219$ to 3 decimal places.

4a

$f(2.2) = \sin 2.2 - \ln 2.2$

$= 0.02003904\ldots,$

$f(2.3) = \sin 2.3 - \ln 2.3$

$= -0.08720391\ldots.$

The values have opposite signs, and the function is continuous on the interval. Hence a root lies between 2.2 and 2.3.

4b

Test the rounding boundaries:

$f(2.2185) = \sin 2.2185 - \ln 2.2185$

$= 0.00064011\ldots,$

$f(2.2195) = \sin 2.2195 - \ln 2.2195$

$= -0.00041430\ldots.$

There is a change of sign. Continuity gives $2.2185 < \alpha < 2.2195$, so $\alpha = 2.219$ to 3 decimal places.

Using Iteration to Approximate a Root Solution 12

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$g(x) = e^{x-1} + x - 6.$

a. Show that $g(x) = 0$ can be rearranged as $x = \ln(6 - x) + 1$.

b. Use $x_{n+1} = \ln(6 - x_n) + 1$, with $x_0 = 2$, to find x_1, x_2, x_3 to 4 decimal places.

Iteration rule

Rearrange $f(x) = 0$ into $x = g(x)$, then use $x_{n+1} = g(x_n)$. We will see why this works shortly.

a

$0 = e^{x-1} + x - 6$

$6 - x = e^{x-1}$

$\ln(6 - x) = x - 1$

$x = \ln(6 - x) + 1.$

Tip

There may be several occurrences of x to isolate. Use the target equation to guide the rearrangement.

Iteration — Calculation and Exam Technique

Solution 13

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$g(x) = e^{x-1} + x - 6$.

a. Show that $g(x) = 0$ can be rearranged as $x = \ln(6 - x) + 1$.

b. Use $x_{n+1} = \ln(6 - x_n) + 1$, with $x_0 = 2$, to find x_1, x_2, x_3 to 4 decimal places.

b

$x_0 = 2, \quad x_1 = \ln(6 - 2) + 1 = 2.3863,$

$x_2 = 2.2847, \quad x_3 = 2.3125 \quad (4 \text{ d.p.}).$

x_0 is the starting value. The later terms are successive approximations to a root.

Calculator tip

Enter 2 first, then enter $\ln(6 - \text{Ans}) + 1$. Press equals repeatedly for successive iterations. Retain unrounded values during the calculation.

Exam tip

Show the substitution for x_1 to earn the method mark. Then state the required values of x_2 and later terms.

Does the Starting Value Matter?

Solution 14

The starting value matters: iteration can approach a different root, or fail to converge. Staircase and cobweb diagrams help explain why.

$f(x) = x^3 - 3x^2 - 2x + 5$.

a. Show there is a root in $3 < x < 4$.

b. Use $x_{n+1} = \sqrt{(x_n^3 - 2x_n + 5)/3}$ to find x_1, x_2, x_3, to 4 d.p., starting from i. $x_0 = 1.5$; ii. $x_0 = 4$.

a

$f(3) = -1, \quad f(4) = 13.$

There is a change of sign, and f is continuous. Hence a root lies in $(3, 4)$.

b(i)

$$x_1 = \sqrt{\frac{1.5^3 - 2(1.5) + 5}{3}} = 1.3385,$$

$x_2 = 1.2545, \quad x_3 = 1.2200.$

The changes become smaller; this sequence appears to converge to a different root.

b(ii)

$x_1 = 4.5092, \quad x_2 = 5.4058,$

$x_3 = 7.1219 \quad (4 \text{ d.p.}).$

The values grow away from the root: this iteration diverges.

Test Your Understanding — Iteration

Solution 15

Edexcel C3 June 2012, Q2

$f(x) = x^3 + 3x^2 + 4x - 12$. The equation $f(x) = 0$ has a single root between 1 and 2.

a. Show that $f(x) = 0$ can be written as $x = \sqrt{4(3-x)/(3+x)}$ for the root between 1 and 2.

b. Use $x_{n+1} = \sqrt{4(3-x_n)/(3+x_n)}$, $x_0 = 1$, to find x_1, x_2, x_3 to 2 d.p.

c. Prove that the root $\alpha = 1.272$ to 3 d.p.

a

$x^3 + 3x^2 = 12 - 4x$

$x^2(x+3) = 4(3-x)$

$x^2 = \dfrac{4(3-x)}{3+x}$.

For the positive root,

$x = \sqrt{\dfrac{4(3-x)}{3+x}}, \qquad x \neq -3.$

b $x_1 = 1.41, \quad x_2 = 1.20, \quad x_3 = 1.31 \quad (2 \text{ d.p.}).$

c

$f(1.2715) = -0.008213624\ldots < 0,$

$f(1.2725) = 0.008272328\ldots > 0.$

Continuity and a sign change give $1.2715 < \alpha < 1.2725$. Thus $\alpha = 1.272$ to 3 d.p.

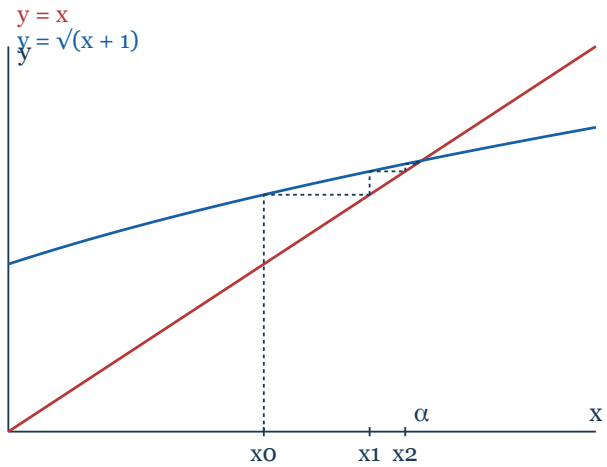
Exam tip

Mark schemes show the minimum acceptable wording. State the endpoint values, change of sign and continuity clearly.

Why Does Iteration Work?

Solution 16

Solve $x^2 - x - 1 = 0$. For the positive root, rearrange as $x = \sqrt{x+1}$ and use $x_{n+1} = \sqrt{x_n + 1}$. Why does this work?



The solution of $x = \sqrt{x+1}$ is the intersection of $y = x$ and $y = \sqrt{x+1}$. Its x-coordinate, α, is a root of the original equation.

With $x_0 = 1$, find $\sqrt{x_0 + 1}$: move vertically to the curve to obtain the y-value.

This gives $x_1 = \sqrt{2} = 1.414\ldots$. Feed it back as the next input: move horizontally to $y = x$, so the y-value becomes the new x-value.

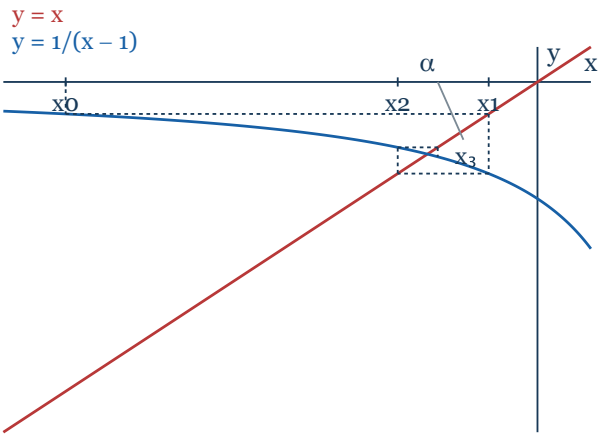
Repeat using $x_1 = 1.414\ldots$ to obtain x_2, and so

The staircase shape gives the name: a staircase diagram. The iterates converge towards α.

Cobweb Diagrams

Solution 17

Another rearrangement of $x^2 - x - 1 = 0$ is $x = 1/(x-1)$. Use $x_{n+1} = 1/(x_n - 1)$. What happens?



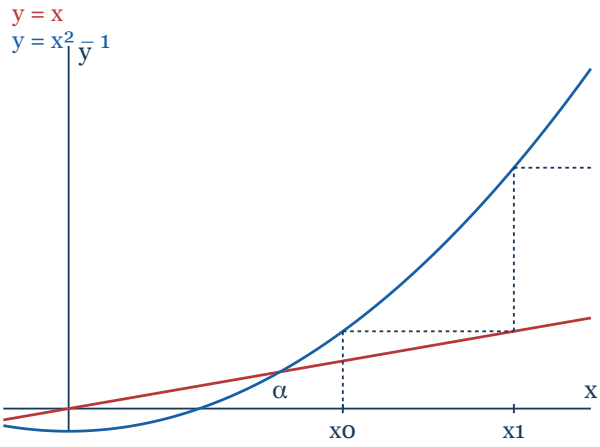
Start with a negative value, as in this diagram. Alternate between the curve $y = 1/(x-1)$ and the line $y = x$.

The terms alternate around α and move closer to it. This is a cobweb diagram. Here iteration converges to the negative root, $\alpha = (1 - \sqrt{5})/2$.

When Iteration Fails

Solution 18

We can also rearrange $x^2 - x - 1 = 0$ as $x = x^2 - 1$, giving $x_{n+1} = x_n^2 - 1$. What happens now?



Start to the right of the positive root. Each iteration moves farther from α, instead of closer. The approximations diverge, so this attempt fails.

Test Your Understanding — Staircase Diagram

Solution 19

$f(x) = x^2 - 8x + 4$.

a. Show that a nonnegative root can be written as $x = \sqrt{8x - 4}$.

b. Starting with $x_0 = 1$, use $x_{n+1} = \sqrt{8x_n - 4}$ to draw a staircase diagram. Label x_0, x_1, x_2 and the root α.

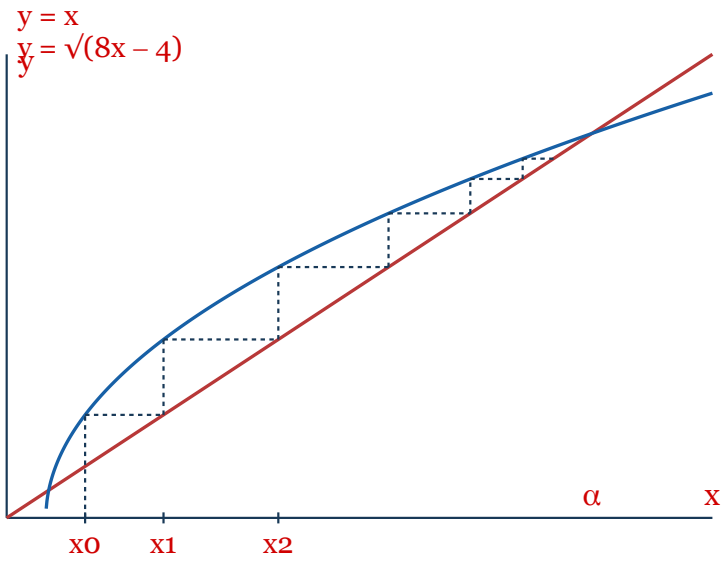
a–b

$x^2 = 8x - 4 \quad \Rightarrow \quad x = \sqrt{8x - 4}$

for a nonnegative root.

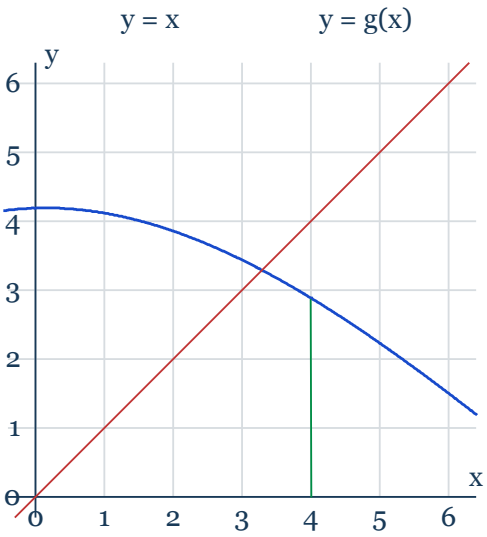
$x_0 = 1,\ x_1 = 2,\ x_2 = \sqrt{12}$. The staircase approaches $\alpha = 4 + 2\sqrt{3}$.

b sketch



Fixed-Point Iteration I

Solution 20

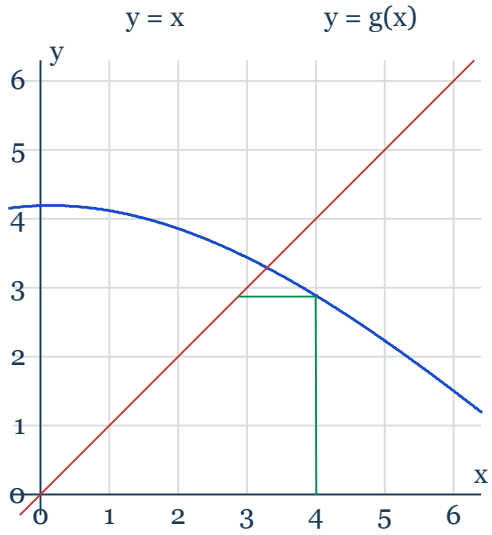


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

1. Mark on $y = g(x)$ the point whose x-coordinate is 4.

Fixed-Point Iteration I

Solution 21

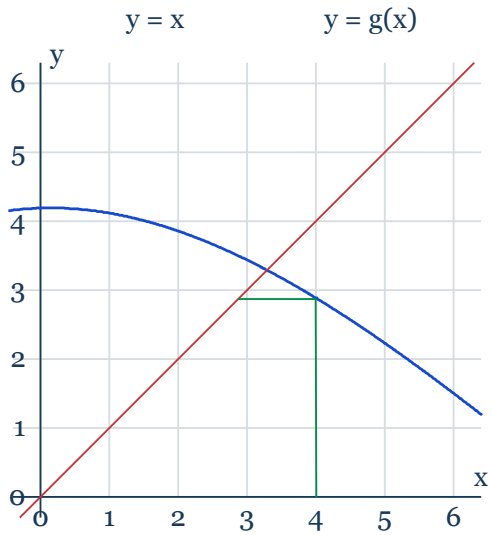


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

2. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration I

Solution 22

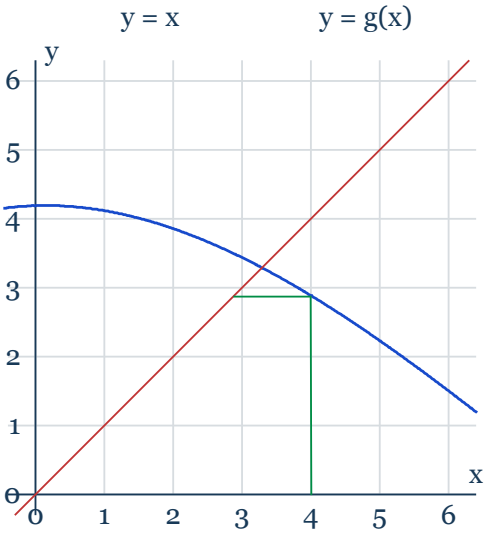


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

3. Draw a vertical line from this point until it reaches the curve $y = g(x)$.

Fixed-Point Iteration I

Solution 23

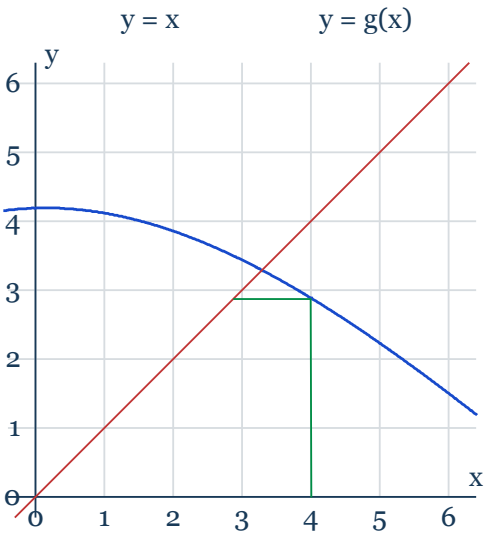


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

4. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration I

Solution 24

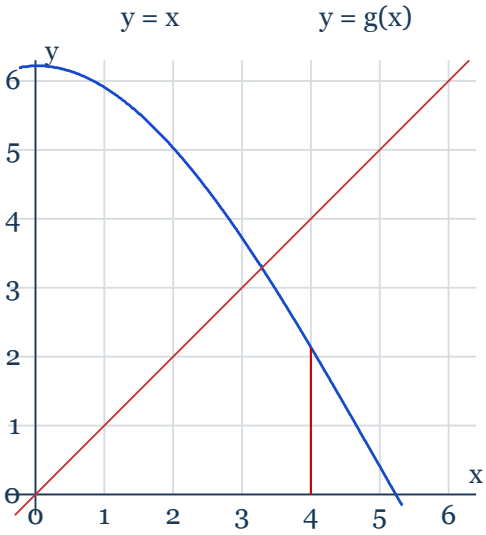


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

5. Repeat steps 3 and 4 several times. What do you notice?

Fixed-Point Iteration II

Solution 25

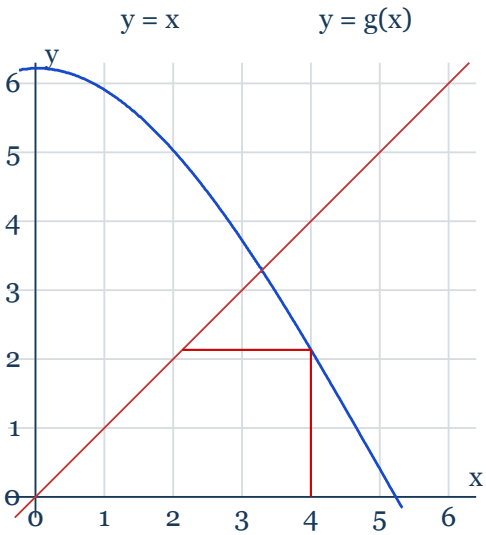


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

1. Mark on $y = g(x)$ the point whose x-coordinate is 4.

Fixed-Point Iteration II

Solution 26

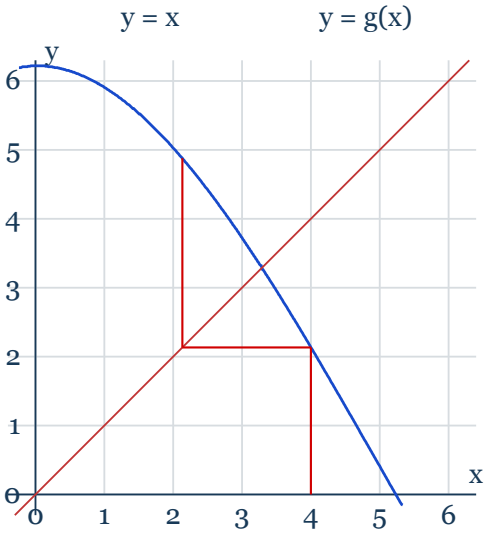


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

2. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration II

Solution 27

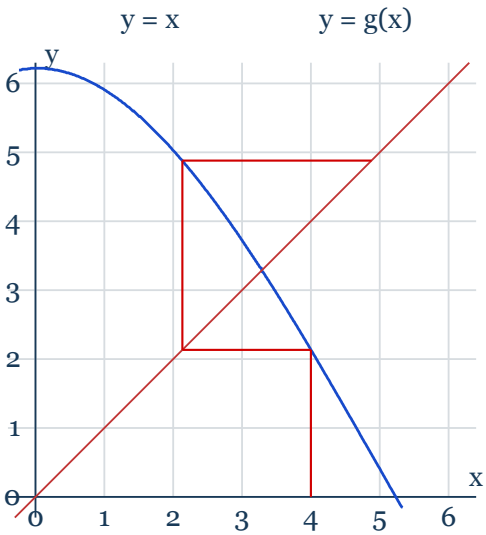


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

3. Draw a vertical line from this point until it reaches the curve $y = g(x)$.

Fixed-Point Iteration II

Solution 28

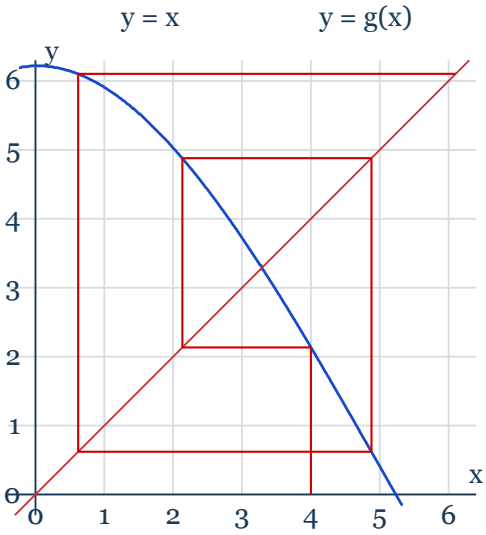


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

4. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration II

Solution 29



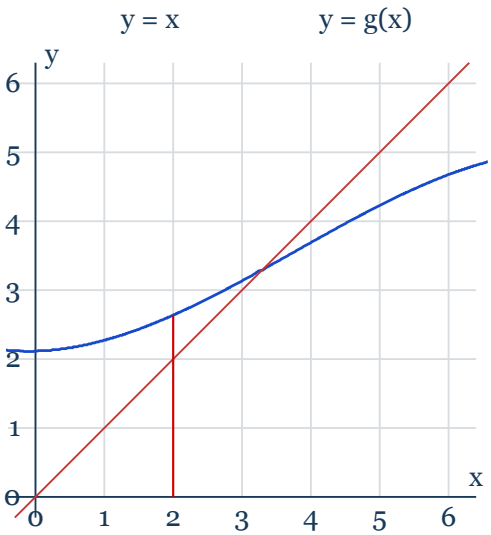
The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

5. Repeat steps 3 and 4 several times. What do you notice?

The cobweb moves away from the fixed point near 3; this starting value does not converge to it.

Fixed-Point Iteration III

Solution 30

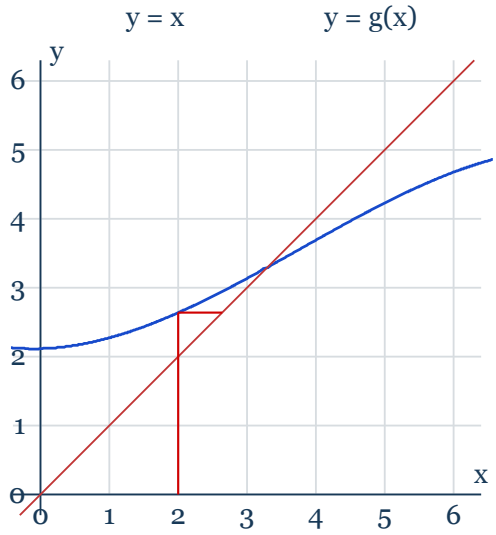


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

1. Mark on $y = g(x)$ the point whose x-coordinate is 2.

Fixed-Point Iteration III

Solution 31

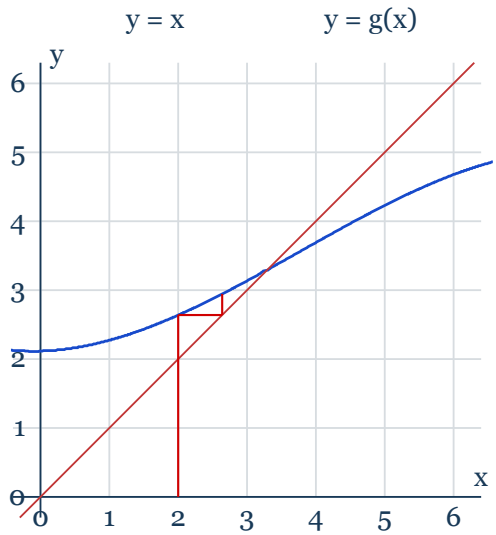


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

2. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration III

Solution 32

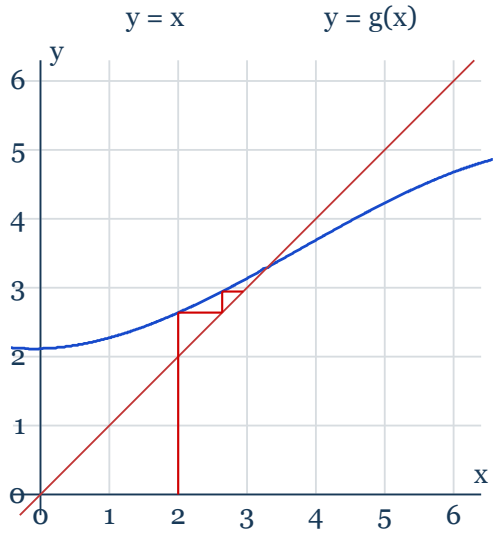


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

3. Draw a vertical line from this point until it reaches the curve $y = g(x)$.

Fixed-Point Iteration III

Solution 33

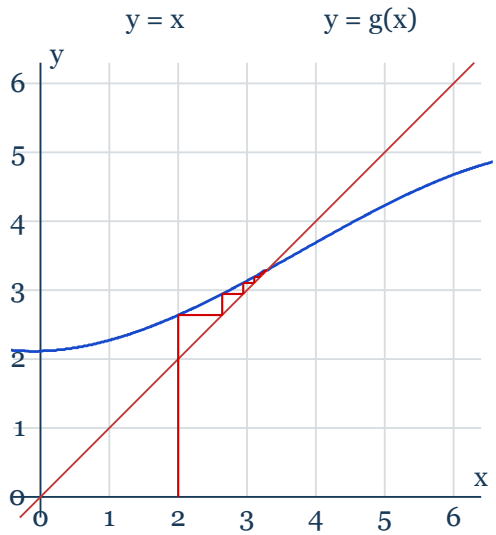


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

4. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration III

Solution 34



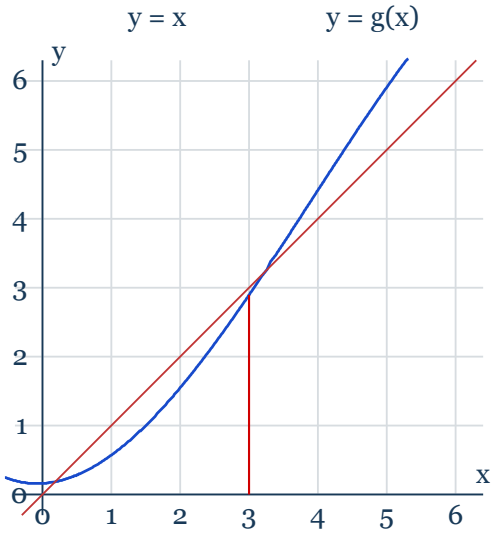
The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

5. Repeat steps 3 and 4 several times. What do you notice?

The staircase approaches the fixed point near 3; the iterations converge to it.

Fixed-Point Iteration IV

Solution 35

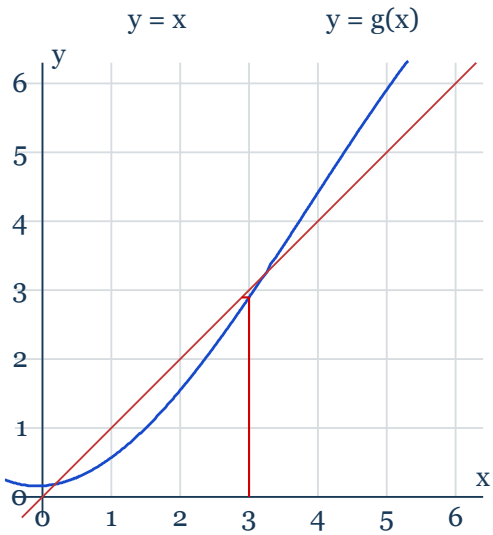


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

1. Mark on $y = g(x)$ the point whose x-coordinate is 3.

Fixed-Point Iteration IV

Solution 36

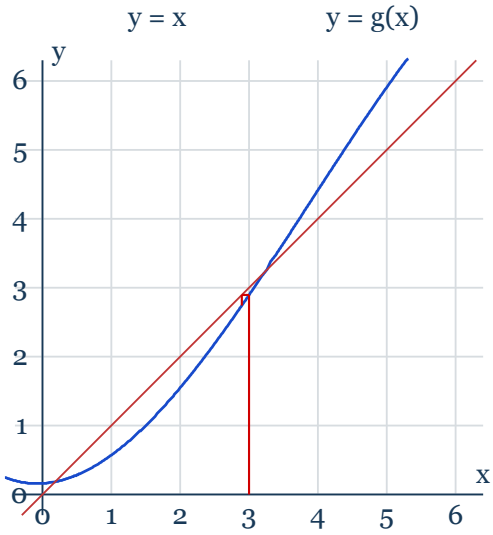


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

2. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration IV

Solution 37

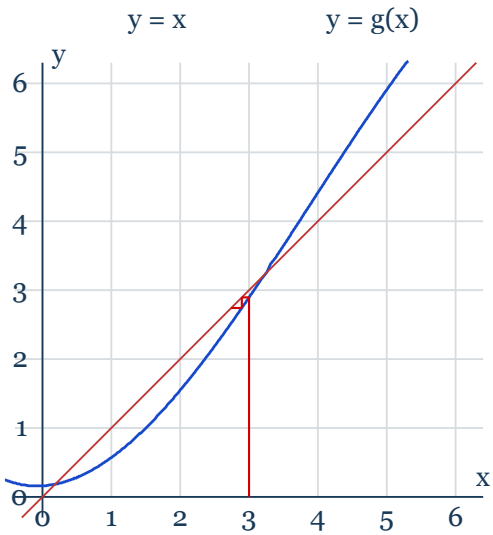


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

3. Draw a vertical line from this point until it reaches the curve $y = g(x)$.

Fixed-Point Iteration IV

Solution 38

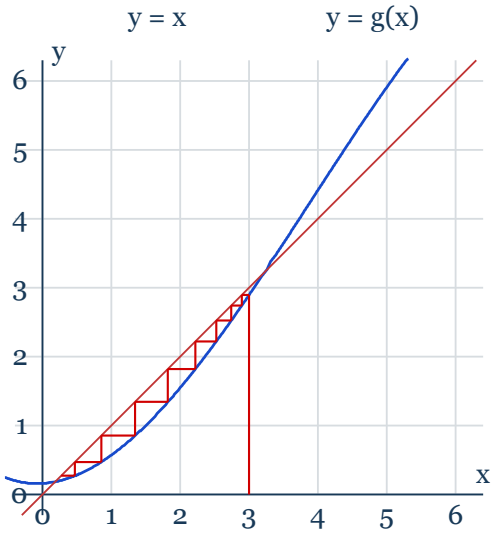


The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

4. Draw a horizontal line from this point until it reaches $y = x$.

Fixed-Point Iteration IV

Solution 39



The graph shows $y = x$ and $y = g(x)$. The line and curve meet at a fixed point whose x-coordinate is close to 3.

5. Repeat steps 3 and 4 several times. What do you notice?

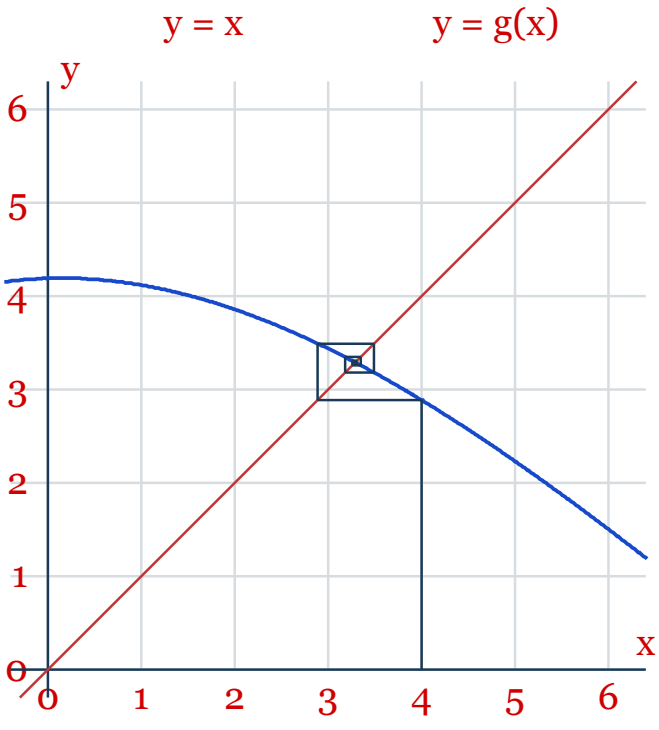
The staircase moves away from the fixed point near 3; this starting value does not converge to it.

Answer — Fixed-Point Iteration I

Solution 40

Describe the construction and its behaviour near the fixed point close to 3.

Diagram



Conclusion

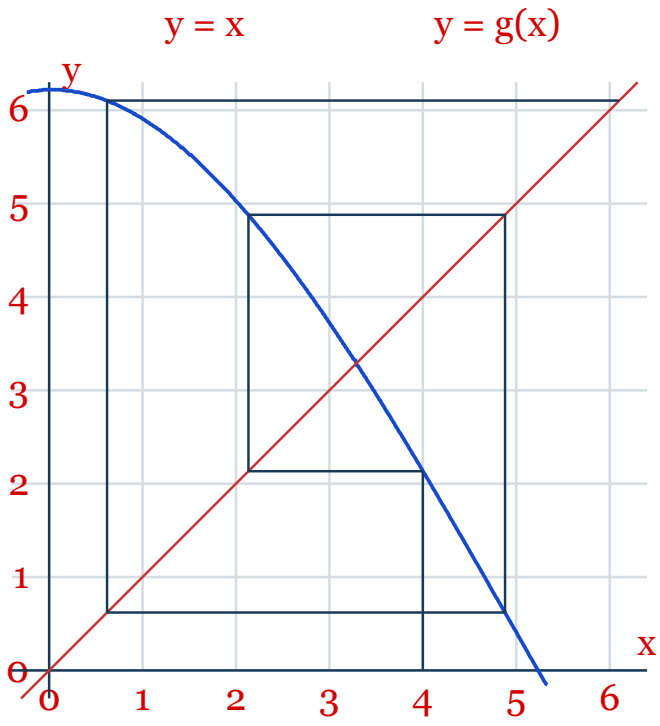
This is a cobweb diagram. The iterations move towards the fixed point close to 3. They converge to that fixed point.

Answer — Fixed-Point Iteration II

Solution 41

Describe the construction and its behaviour near the fixed point close to 3.

Diagram



Conclusion

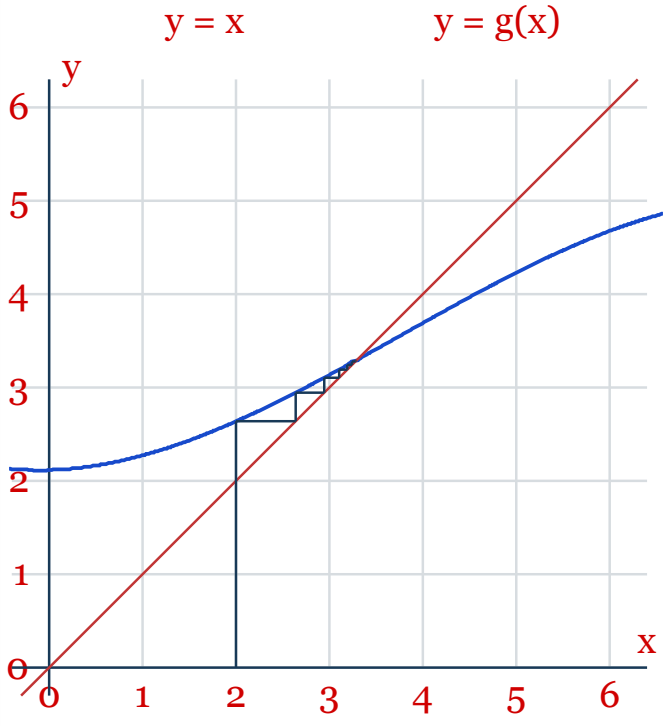
This is a cobweb diagram. The iterations move away from the fixed point close to 3. This starting value does not converge to that fixed point.

Answer — Fixed-Point Iteration III

Solution 42

Describe the construction and its behaviour near the fixed point close to 3.

Diagram



Conclusion

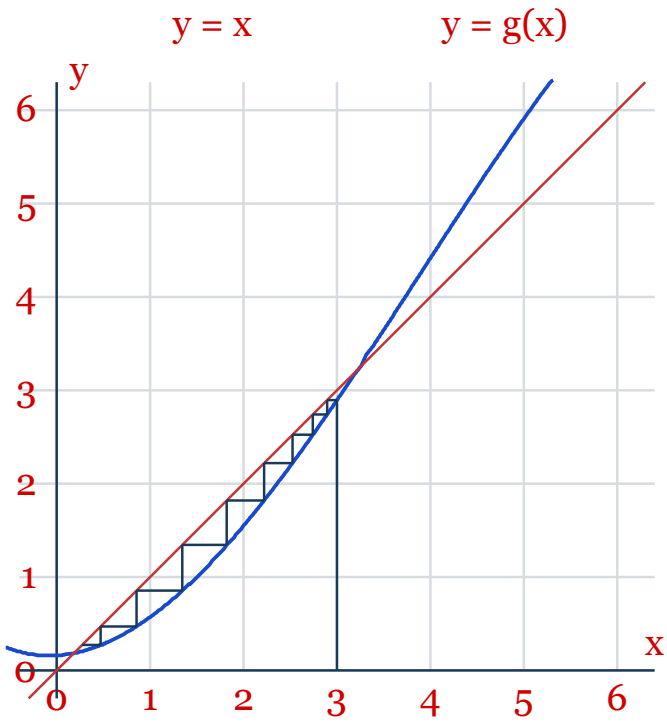
This is a staircase diagram. The iterations move towards the fixed point close to 3. They converge to that fixed point.

Answer — Fixed-Point Iteration IV

Solution 43

Describe the construction and its behaviour near the fixed point close to 3.

Diagram



Conclusion

This is a staircase diagram. The iterations move away from the fixed point close to 3. This starting value does not converge to that fixed point.

Exercise 10B — Pages 280–282

Solution 44

3. $f(x) = x^2 - 6x + 1$.

a. Show that $f(x) = 0$ can be written as $x = \sqrt{6x - 1}$.

b. Sketch $y = x$ and $y = \sqrt{6x - 1}$.

c. State the number of roots.

d. Explain graphically why $x_{n+1} = \sqrt{6x_n - 1}$, $x_0 = 2$, converges to a root.

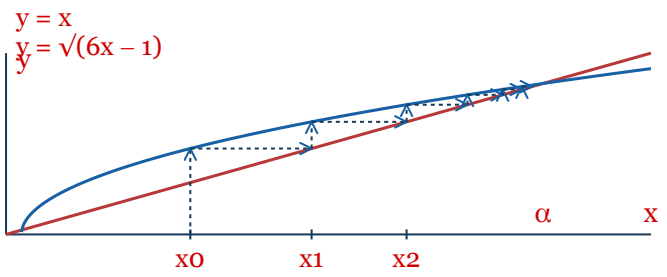
e. By sketching a diagram, explain why $x_{n+1} = (x_n^2 + 1)/6$, $x_0 = 10$, moves away from the roots.

3a

$x^2 - 6x + 1 = 0 \Rightarrow x^2 = 6x - 1$.

Both roots are positive, so take the positive square root: $x = \sqrt{6x - 1}$.

3b



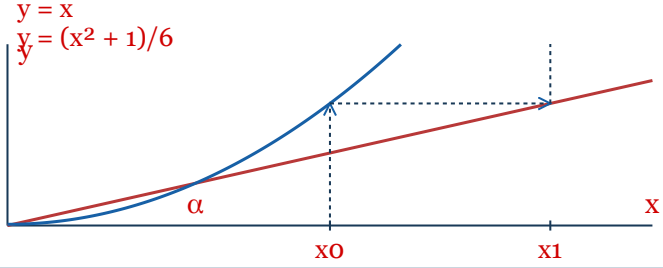
3c

The curve meets the line twice: there are two roots.

3d

Starting at 2, the staircase approaches the larger fixed point $3 + 2\sqrt{2}$. Hence the iteration converges to this root.

3e



3e conclusion

Starting at 10, the steps move to larger values, farther from the fixed points. This iteration diverges.

Exercise 10B — Pages 280–282 Solution 45

6. $f(x) = x^4 - 3x^3 - 6$.

a. Rearrange $f(x) = 0$ as $x = \sqrt[3]{px^4 + q}$, finding p, q.

b. With $x_0 = 0$, use $x_{n+1} = \sqrt[3]{px_n^4 + q}$ to find x_1, x_2, x_3 to 3 d.p.

c. Prove that the negative root $\alpha = -1.132$ to 3 d.p.

6a

$x^4 - 3x^3 - 6 = 0$

$3x^3 = x^4 - 6$

$\frac{3x^3}{3} = \frac{x^4}{3} - \frac{6}{3}$

$x^3 = \frac{x^4}{3} - 2$

$x = \sqrt[3]{\frac{x^4}{3} - 2}.$

Hence $p = 1/3,\ q = -2$.

6b

$x_1 = \sqrt[3]{\frac{1}{3}(0)^4 - 2}$

$= -1.259921\ldots$

$x_2 = \sqrt[3]{\frac{(-1.259921\ldots)^4}{3} - 2}$

$= -1.050733\ldots$

$x_3 = \sqrt[3]{\frac{(-1.050733\ldots)^4}{3} - 2}$

$= -1.168069\ldots.$

To 3 d.p., $x_1 = -1.260$, $x_2 = -1.051$, $x_3 = -1.168$.

6c

$f(-1.1315) = (-1.1315)^4 - 3(-1.1315)^3 - 6$

$= -0.0148\ldots < 0,$

$f(-1.1325) = (-1.1325)^4 - 3(-1.1325)^3 - 6$

$= 0.0024\ldots > 0.$

The polynomial is continuous and changes sign, so $-1.1325 < \alpha < -1.1315$. Thus $\alpha = -1.132$ to 3 d.p.

Introduction to Newton–Raphson Solution 46

Pure Mathematics Year 2 · Chapter 10

Numerical methods

Introduction to the Newton–Raphson method

Name the movie.



Solving a Cubic Numerically

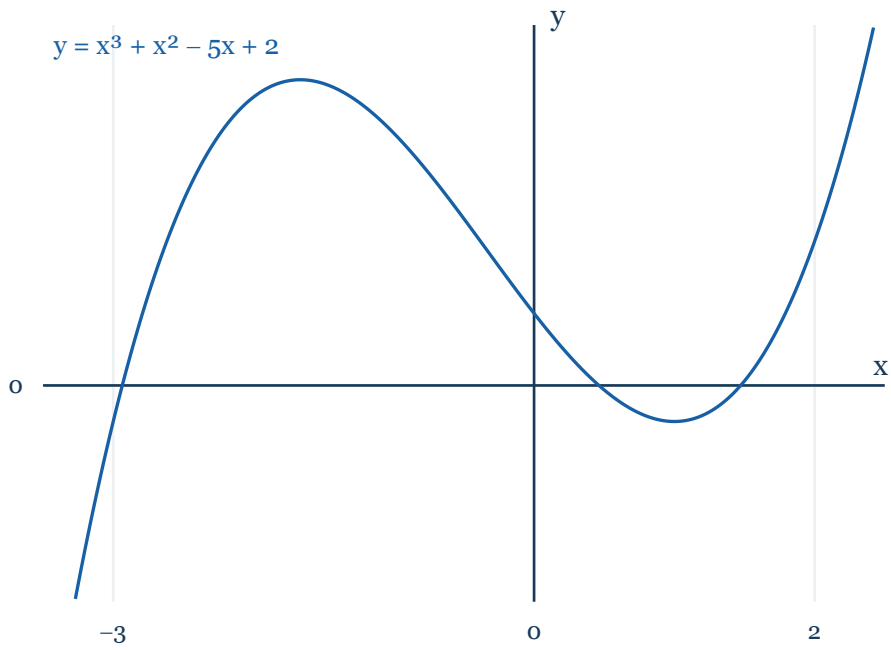
Solution 47

Use the Newton–Raphson method to solve
$f(x) = x^3 + x^2 - 5x + 2 = 0$.

Three Roots of the Cubic

Solution 48

$x^3 + x^2 - 5x + 2 = 0$

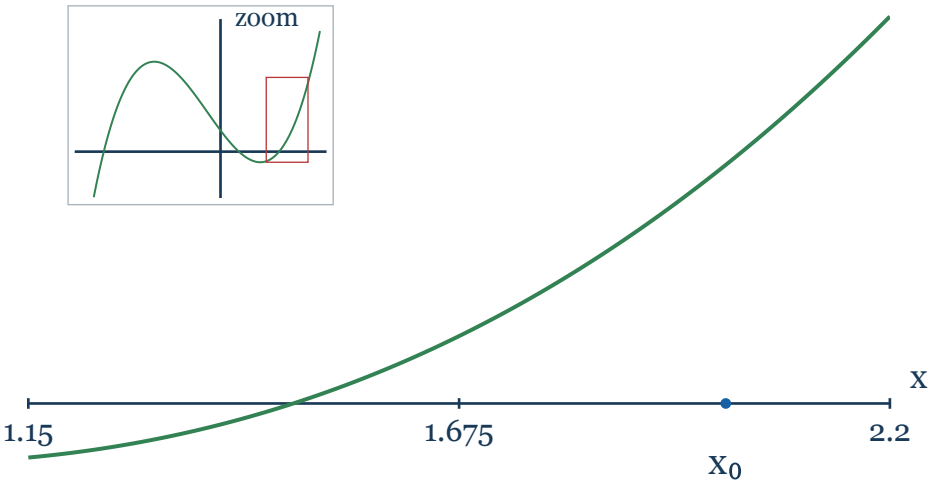


Zoom in on One of the Roots

Solution 49

$x^3 + x^2 - 5x + 2 = 0$

Zoom in on the right-hand root. Start with $x_0 = 2$.

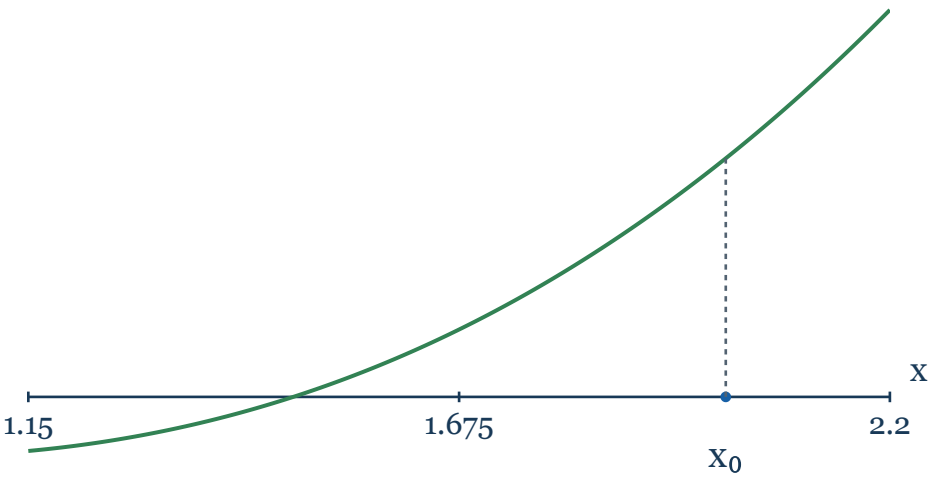


Zoom in on One of the Roots

Solution 50

$x^3 + x^2 - 5x + 2 = 0$

Zoom in on the right-hand root. Start with $x_0 = 2$.

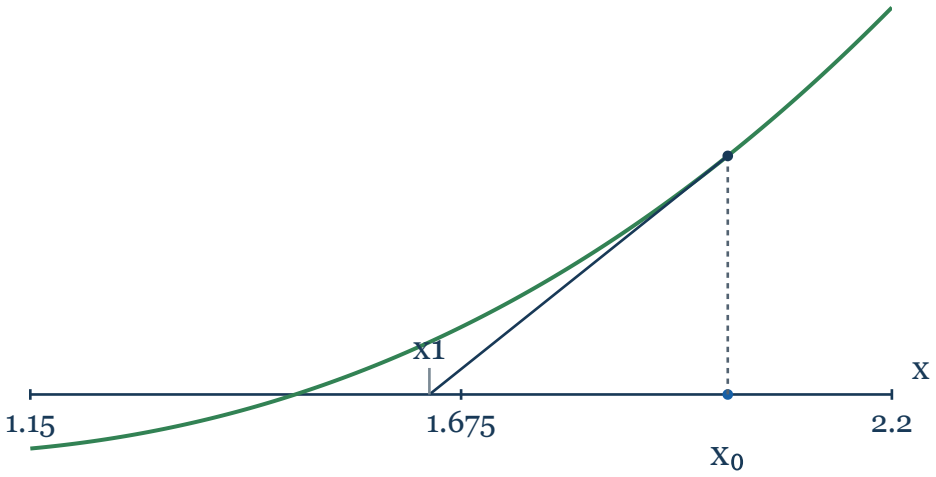


Zoom in on One of the Roots

Solution 51

$x^3 + x^2 - 5x + 2 = 0$

Zoom in on the right-hand root. Start with $x_0 = 2$.

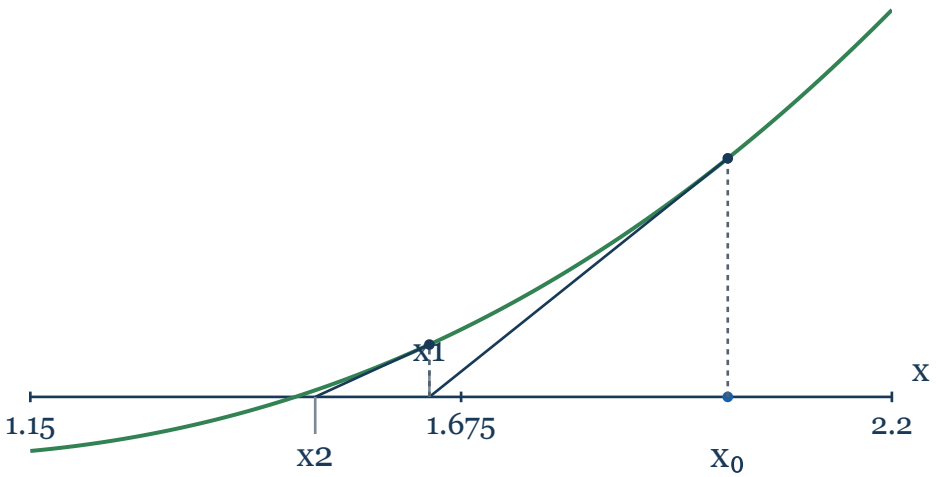


Zoom in on One of the Roots

Solution 52

$x^3 + x^2 - 5x + 2 = 0$

Zoom in on the right-hand root. Start with $x_0 = 2$.

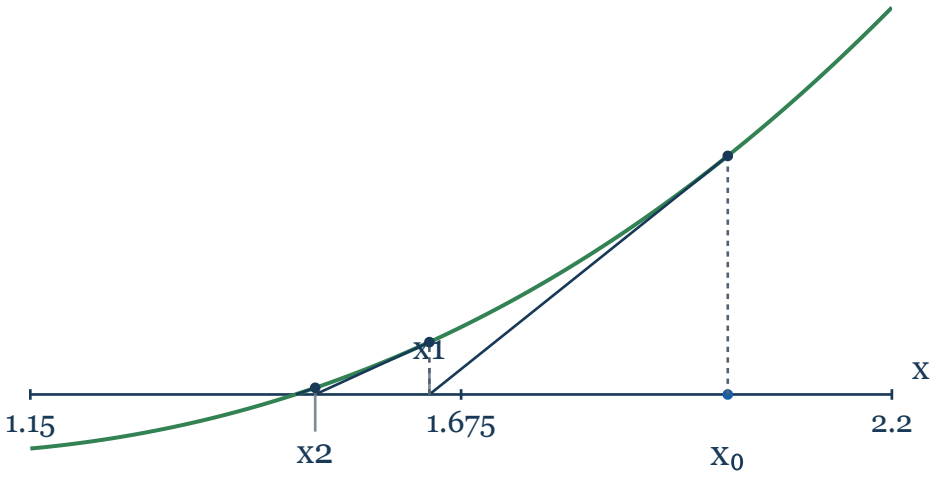


Zoom in on One of the Roots

Solution 53

$x^3 + x^2 - 5x + 2 = 0$

Zoom in on the right-hand root. Start with $x_0 = 2$.

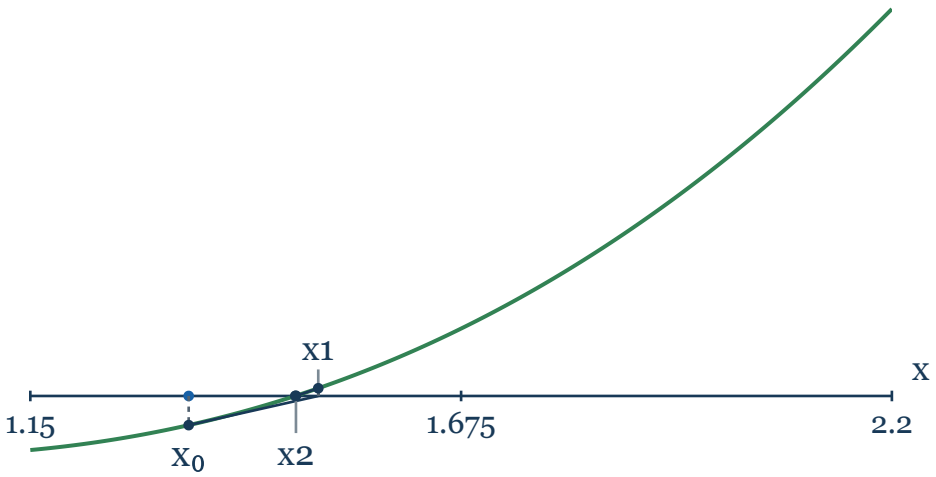


What Happens If We Start from the Left?

Solution 54

$f(x) = x^3 + x^2 - 5x + 2$

Try a starting point to the left of the right-hand root.

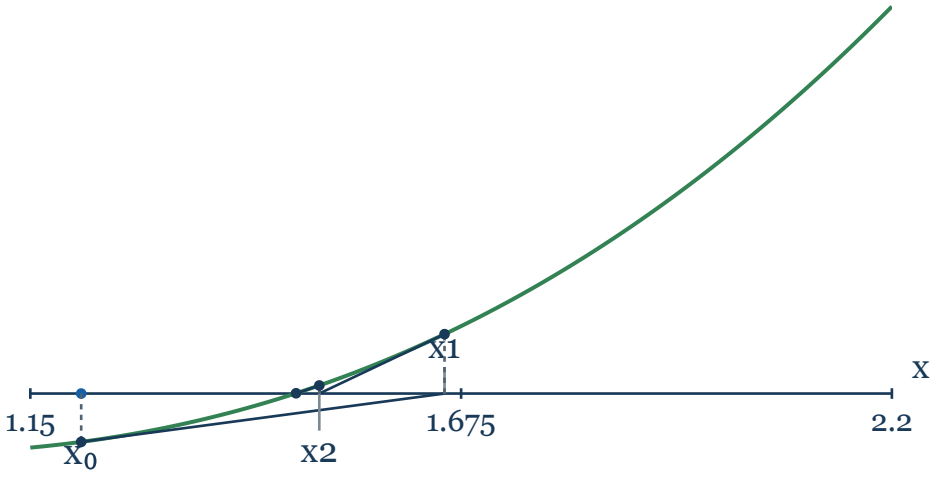


What Happens If We Start from the Left?

Solution 55

$f(x) = x^3 + x^2 - 5x + 2$

Try a starting point to the left of the right-hand root.

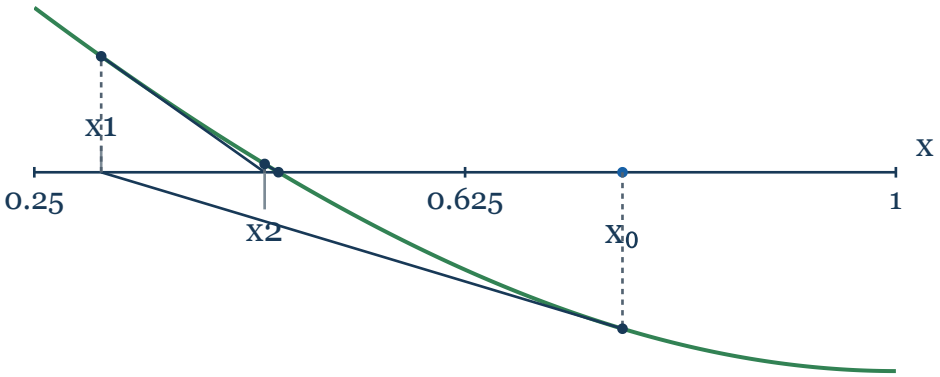


What Happens If We Start from the Left?

Solution 56

$f(x) = x^3 + x^2 - 5x + 2$

Try a starting point near the middle root.

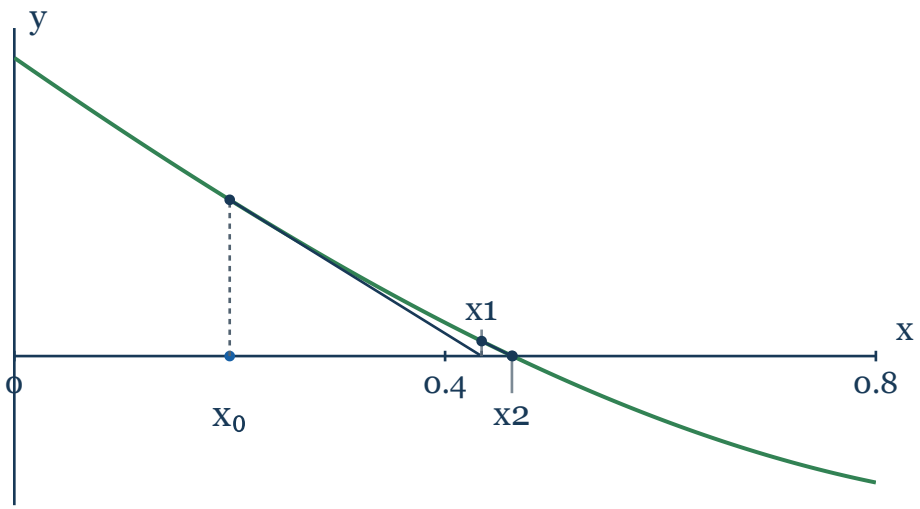


What Happens If We Start from the Left?

Solution 57

$f(x) = x^3 + x^2 - 5x + 2$

Try a starting point near the middle root.

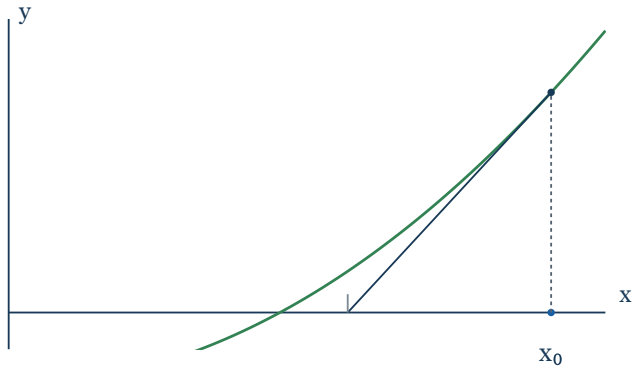


Deriving the Newton–Raphson Formula

Solution 58

Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.

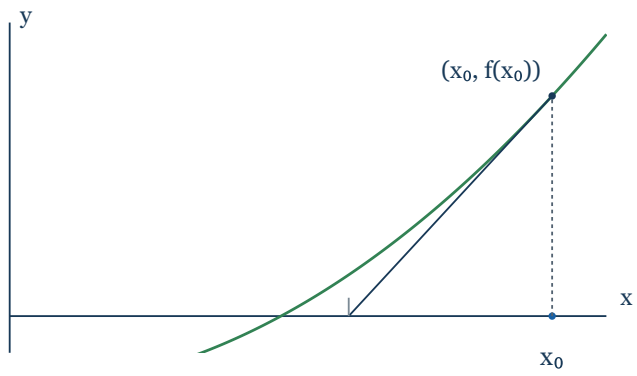


Deriving the Newton–Raphson Formula

Solution 59

Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.



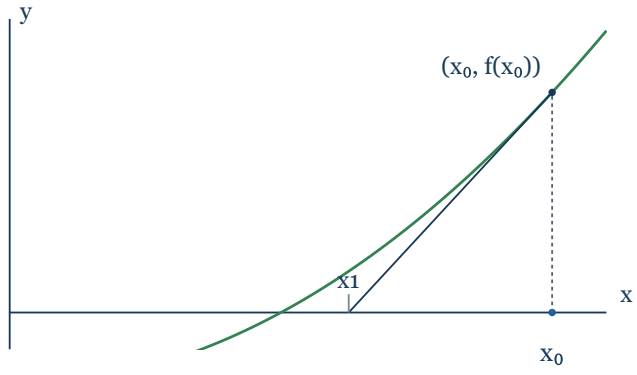
Deriving the Newton–Raphson Formula

Solution 60

Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.

x_1 is where the tangent crosses the x-axis.



Deriving the Newton–Raphson Formula

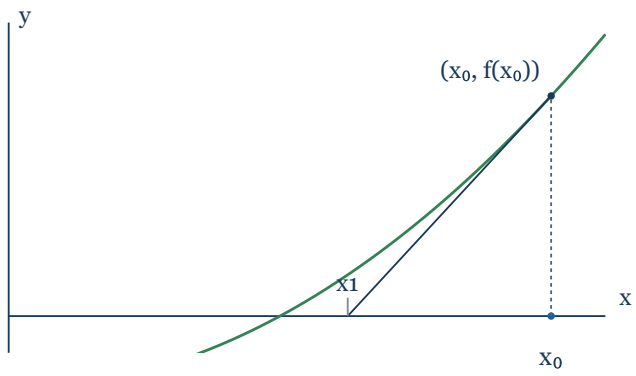
Solution 61

Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.

x_1 is where the tangent crosses the x-axis.

Gradient of tangent $= \dfrac{f(x_0)}{x_0 - x_1}$.



Deriving the Newton–Raphson Formula

Solution 62

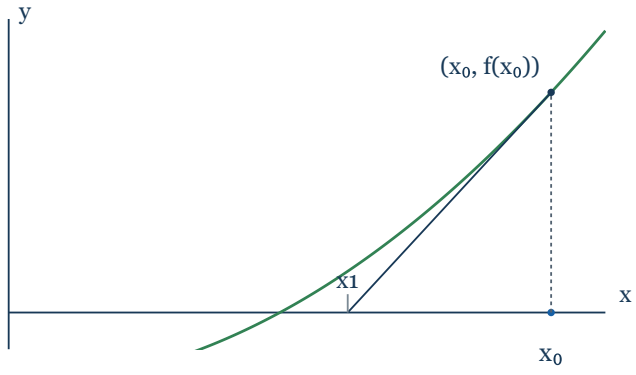
Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.

x_1 is where the tangent crosses the x-axis.

Gradient of tangent $= \dfrac{f(x_0)}{x_0 - x_1}$.

$f'(x_0) = \dfrac{f(x_0)}{x_0 - x_1}$.



Deriving the Newton–Raphson Formula

Solution 63

Example — solve $f(x) = 0$.

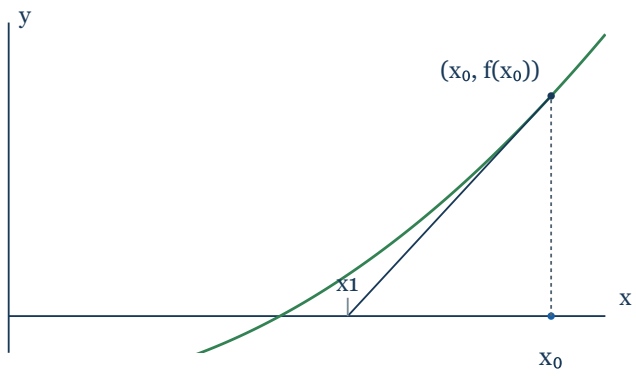
Set x_0 and look at the point $(x_0, f(x_0))$.

x_1 is where the tangent crosses the x-axis.

Gradient of tangent $= \dfrac{f(x_0)}{x_0 - x_1}$.

$f'(x_0) = \dfrac{f(x_0)}{x_0 - x_1}$.

$f(x_0) = (x_0 - x_1)f'(x_0)$.



Deriving the Newton–Raphson Formula

Solution 64

Example — solve $f(x) = 0$.

Set x_0 and look at the point $(x_0, f(x_0))$.

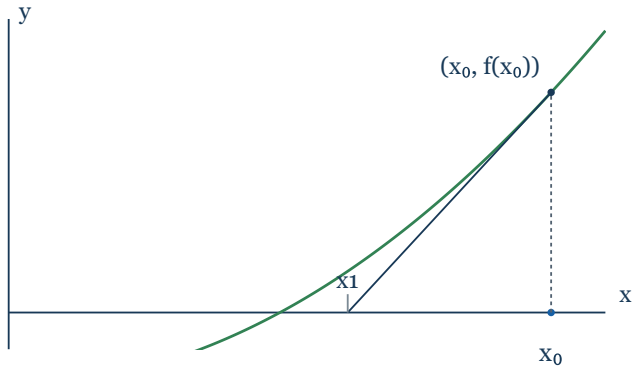
x_1 is where the tangent crosses the x-axis.

Gradient of tangent $= \dfrac{f(x_0)}{x_0 - x_1}$.

$f'(x_0) = \dfrac{f(x_0)}{x_0 - x_1}$.

$f(x_0) = (x_0 - x_1)f'(x_0)$.

$x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)}$.



Repeat the process to obtain the iterative formula.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

The function and its derivative must be defined, with $f'(x_n) \neq 0$.

Repeat the process to obtain the iterative formula.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Calculator hint

Save $f(x)$ as Y1 in Graph mode. Use the Vars key to access Y1 in Run mode.

The iterative formula is

$$\text{Ans} - \frac{\text{Y1(Ans)}}{\dfrac{d}{dx}\text{Y1}(x)\Big|_{x=\text{Ans}}}.$$

Applying Newton–Raphson to the Cubic

Solution 67

Use the Newton–Raphson method to solve
$f(x) = x^3 + x^2 - 5x + 2 = 0$.

Applying Newton–Raphson to the Cubic

Solution 68

Use the Newton–Raphson method to solve
$f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

Applying Newton–Raphson to the Cubic

Solution 69

Use the Newton–Raphson method to solve
$f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$.

Applying Newton–Raphson to the Cubic

Solution 70

Use the Newton–Raphson method to solve
$f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$.

$x_{n+1} = x_n - \dfrac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}$.

Applying Newton–Raphson to the Cubic

Solution 71

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Applying Newton–Raphson to the Cubic

Solution 72

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

Applying Newton–Raphson to the Cubic

Solution 73

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

Applying Newton–Raphson to the Cubic

Solution 74

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

$$x_2 = 1.497200048\ldots$$

Applying Newton–Raphson to the Cubic

Solution 75

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

$$x_2 = 1.497200048\ldots$$

Applying Newton–Raphson to the Cubic

Solution 76

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

$$x_2 = 1.497200048\ldots$$

$$x_3 = 1.473521722\ldots$$

Applying Newton–Raphson to the Cubic

Solution 77

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

$$x_2 = 1.497200048\ldots$$

$$x_3 = 1.473521722\ldots$$

$$x_4 = 1.472834484\ldots, \quad x_5 = 1.472833909\ldots$$

Applying Newton–Raphson to the Cubic

Solution 78

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$f'(x) = 3x^2 + 2x - 5$.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Using $x_0 = 2$.

Calculator hint

Enter the recurrence using Ans in place of x_n, then press equals repeatedly.

$$x_1 = \frac{18}{11} = 1.636363636\ldots$$

$$x_2 = 1.497200048\ldots$$

$$x_3 = 1.473521722\ldots$$

$$x_4 = 1.472834484\ldots, \quad x_5 = 1.472833909\ldots$$

This suggests that one root is 1.4728, correct to 4 decimal places. We still need to verify the rounding.

Verify the Root to Four Decimal Places

Solution 79

Show that the root is 1.4728, correct to 4 decimal places.

$f(x) = x^3 + x^2 - 5x + 2$.

Verify the Root to Four Decimal Places

Solution 80

Show that the root is 1.4728, correct to 4 decimal places.

$f(x) = x^3 + x^2 - 5x + 2$.

The rounding boundaries are 1.47275 and 1.47285.

Verify the Root to Four Decimal Places

Solution 81

Show that the root is 1.4728, correct to 4 decimal places.

$f(x) = x^3 + x^2 - 5x + 2.$

The rounding boundaries are 1.47275 and 1.47285.

$f(1.47275) = 1.47275^3 + 1.47275^2 - 5(1.47275) + 2$
$= -3.73641078\ldots \times 10^{-4} < 0.$

Verify the Root to Four Decimal Places

Solution 82

Show that the root is 1.4728, correct to 4 decimal places.

$f(x) = x^3 + x^2 - 5x + 2.$

The rounding boundaries are 1.47275 and 1.47285.

$f(1.47275) = 1.47275^3 + 1.47275^2 - 5(1.47275) + 2$
$= -3.73641078\ldots \times 10^{-4} < 0.$

$f(1.47285) = 1.47285^3 + 1.47285^2 - 5(1.47285) + 2$
$= 7.166\ldots \times 10^{-5} > 0.$

Verify the Root to Four Decimal Places

Solution 83

Show that the root is 1.4728, correct to 4 decimal places.

$f(x) = x^3 + x^2 - 5x + 2$.

The rounding boundaries are 1.47275 and 1.47285.

$f(1.47275) = 1.47275^3 + 1.47275^2 - 5(1.47275) + 2$
$= -3.73641078\ldots \times 10^{-4} < 0.$

$f(1.47285) = 1.47285^3 + 1.47285^2 - 5(1.47285) + 2$
$= 7.166\ldots \times 10^{-5} > 0.$

There is a change of sign. Since f is continuous, there is a root in $(1.47275, 1.47285)$. It therefore rounds to 1.4728 to 4 decimal places.

Find the Other Roots

Solution 84

Use the Newton–Raphson method to solve $f(x) = x^3 + x^2 - 5x + 2 = 0$.

$$x_{n+1} = x_n - \frac{x_n^3 + x_n^2 - 5x_n + 2}{3x_n^2 + 2x_n - 5}.$$

Choose a different starting value to find another root: use $x_0 = 0$.

Spreadsheet setup

Column A: x. Column B: $f(x)$. Column C: $f'(x)$.

Enter 0 in A2. In B2 enter:

```
=A2^3+A2^2-5*A2+2
```

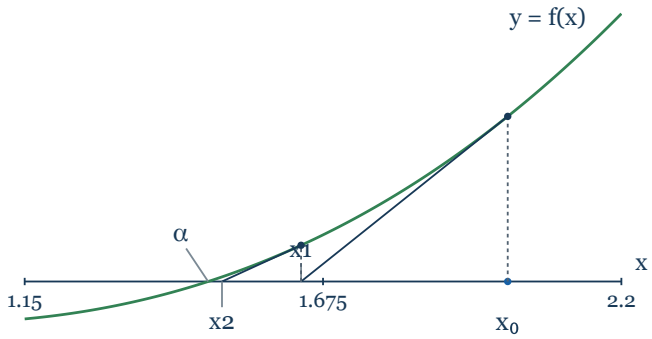
The Newton–Raphson Process

Solution 85

Also known as Newton's method. We wish to find the root α. Start with an approximation x_0, which may be well off the mark.

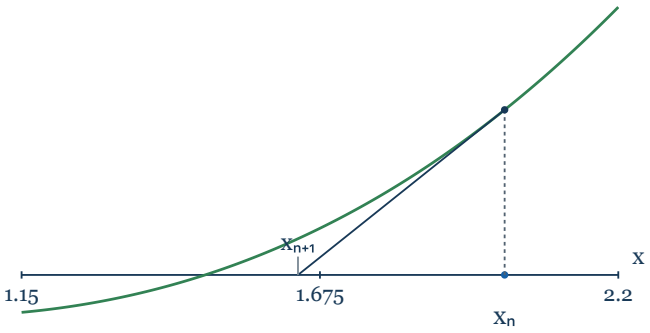
Follow the direction of the curve using its tangent. If the curve were reasonably straight, the point where the tangent hits the x-axis would be close to the root.

Repeat to obtain, hopefully, increasingly accurate approximations. Can you find a formula for x_{n+1} in terms of x_n?



Deriving the Formula from the Tangent

Solution 86



At $x = x_n$, the tangent has gradient $f'(x_n)$.

Using Year 1 coordinate geometry:

$$y - f(x_n) = f'(x_n)(x - x_n).$$

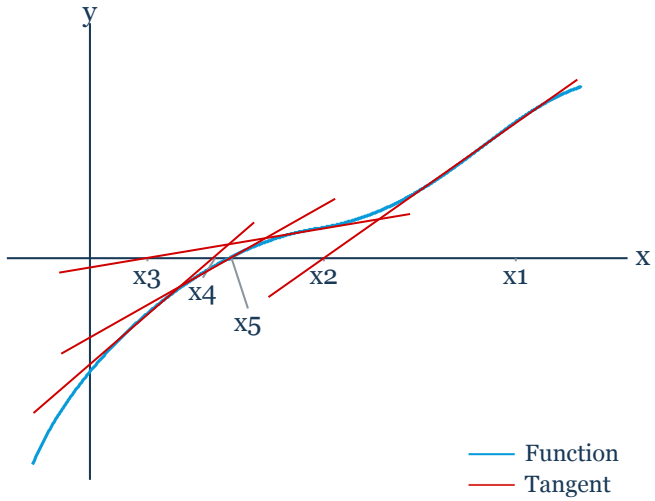
At the tangent's intersection with the axis, $y = 0$ and $x = x_{n+1}$.

$$-f(x_n) = f'(x_n)(x_{n+1} - x_n).$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Newton–Raphson — Animated Demonstration

Solution 87



Follow each step using Next or Space. The original demonstration starts at x_1.

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

$$x_2 = x_1 - f(x_1)/f'(x_1)$$
$$x_3 = x_2 - f(x_2)/f'(x_2)$$
$$x_4 = x_3 - f(x_3)/f'(x_3)$$
$$x_5 = x_4 - f(x_4)/f'(x_4)$$

Demonstration supplied in the source slides, credited there to Wikipedia; reconstructed as native geometry.

Example — Solving x = cos x

Solution 88

Return to $x = \cos x$, with $x_0 = 0.5$. Use radians: the differentiation formula assumes radian measure.

Recurrence

$$f(x) = x - \cos x, \quad f'(x) = 1 + \sin x.$$

This writes the equation in the form $f(x) = 0$.

$$x_{n+1} = x_n - \frac{x_n - \cos x_n}{1 + \sin x_n}.$$

Iterations

$$x_1 = 0.5 - \frac{0.5 - \cos 0.5}{1 + \sin 0.5} = 0.7552224171\ldots$$

$$x_2 = 0.7391416661\ldots$$

$$x_3 = 0.7390851339\ldots$$

After three iterations the approximation agrees with the root to 7 decimal places.

Calculator tip

Enter 0.5 and press equals. Then enter

$$\text{Ans} - \frac{\text{Ans} - \cos(\text{Ans})}{1 + \sin(\text{Ans})}$$

and press equals repeatedly.

Quickfire Questions

Solution 89

Using Newton–Raphson, state the recurrence for each function.

a. $f(x) = x^3 - 2$

a

$$x_{n+1} = x_n - \frac{x_n^3 - 2}{3x_n^2}.$$

b. $f(x) = \tan x$. Use radians; $x_n \neq \pi/2 + k\pi$, $k \in \mathbb{Z}$.

b

$$x_{n+1} = x_n - \frac{\tan x_n}{\sec^2 x_n}$$
$$= x_n - \frac{1}{2}\sin(2x_n).$$

c. $f(x) = x^2 - x - 1$

c

$$x_{n+1} = x_n - \frac{x_n^2 - x_n - 1}{2x_n - 1}.$$

Example Exam Question

Solution 90

Edexcel FP1 June 2013(R), Q3c

$f(x) = \frac{1}{2}x^4 - x^3 + x - 3$. The equation $f(x) = 0$ has a root β in $[-2, -1]$.

c. Starting with -1.5, apply Newton–Raphson once to obtain a second approximation to β. Give your answer to 2 decimal places. (5 marks)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

c

$$f'(x) = 2x^3 - 3x^2 + 1.$$

$$x_{n+1} = x_n - \frac{\frac{1}{2}x_n^4 - x_n^3 + x_n - 3}{2x_n^3 - 3x_n^2 + 1}.$$

$$\beta_1 = -1.5 - \frac{1.40625}{-12.5} = -1.3875.$$

The second approximation is $\boxed{-1.39}$ to 2 d.p.

Test Your Understanding

Solution 91

Edexcel FP1 January 2010, Q2c

$f(x) = 3x^2 - 11/x^2$.

c. Starting with 1.4, apply Newton–Raphson once to obtain a second approximation to α. Give your answer to 3 decimal places. (5 marks)

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

c

$f'(x) = 6x + 22x^{-3}$.

$f(1.4) = 0.267755\ldots, \quad f'(1.4) = 16.417492\ldots$

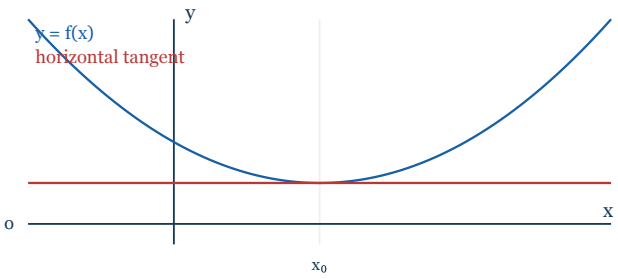
$$x_1 = 1.4 - \frac{f(1.4)}{f'(1.4)}$$

$$= 1.4 - \frac{0.267755\ldots}{16.417492\ldots} = 1.383690\ldots$$

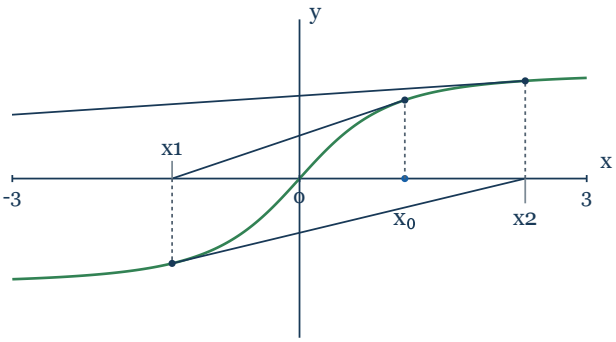
Hence $\boxed{x_1 = 1.384}$ to 3 d.p.

When Does Newton–Raphson Fail?

Solution 92



If the starting value is a stationary point, $f'(x_0) = 0$. Newton–Raphson divides by zero. For a stationary point away from the axis, the horizontal tangent never reaches the axis.



For $f(x) = x/\sqrt{1+x^2}$, the iterates can oscillate on opposite sides of $\alpha = 0$ and move farther away. Newton–Raphson is an iteration and need not converge.

Exercise 10C — Pages 284–285 Solution 93

1. $f(x) = x^3 - 2x - 1$.

a. Show a root α lies in $1 < \alpha < 2$.

b. Starting with $x_0 = 1.5$, apply Newton–Raphson once. Give the second approximation to 3 d.p.

2. $f(x) = x^2 - 4/x + 6x - 10$, $x \neq 0$.

a. Find $f'(x)$. (2 marks)

A root lies in $[-0.4, -0.3]$.

b. Starting with -0.4, apply Newton–Raphson once, giving the second approximation to 3 d.p. (4 marks)

1a

$f(1) = -2, \quad f(2) = 3.$

Continuity and the sign change give a root in $(1, 2)$.

2a

$$f'(x) = 2x + \frac{4}{x^2} + 6 = 2\left(x + \frac{2}{x^2} + 3\right).$$

1b

$f'(x) = 3x^2 - 2.$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 1.5 - \frac{-0.625}{4.75} = 1.631578\ldots$$

Hence $x_1 = 1.632$ to 3 d.p.

2b

$$x_1 = -0.4 - \frac{(-0.4)^2 - 4/(-0.4) + 6(-0.4) - 10}{2(-0.4) + 4/(-0.4)^2 + 6}$$

$$= -0.4 - \frac{-2.24}{30.2} = -0.325827\ldots$$

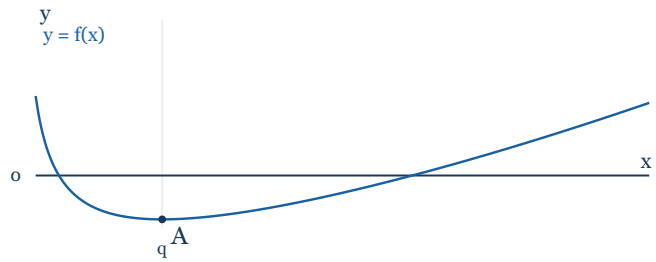
Hence $x_1 = -0.326$ to 3 d.p.

Exercise 10C — Pages 284–285 Solution 94

3. $f(x) = x^{3/2} - e^{-x} + 1/\sqrt{x} - 2$, $x > 0$. The diagram shows $y = f(x)$. Point A, with x-coordinate q, is stationary. A root α lies in $[1.2, 1.3]$.

a. Explain why $x_0 = q$ is unsuitable. (1 mark)

b. Starting with $x_0 = 1.2$, apply Newton–Raphson once. Give the second approximation to 3 d.p. (4 marks)



3a

At the stationary point, $f'(q) = 0$. The Newton–Raphson formula would divide by zero, so this starting value cannot be used.

3b

$$f'(x) = \tfrac{3}{2}\sqrt{x} + e^{-x} - \frac{1}{2x^{3/2}}.$$

$$f(1.2) = 1.2^{3/2} - e^{-1.2} + 1/\sqrt{1.2} - 2$$

$$= -0.073789\ldots$$

$$f'(1.2) = \tfrac{3}{2}\sqrt{1.2} + e^{-1.2} - \frac{1}{2(1.2)^{3/2}}$$

$$= 1.563999\ldots$$

$$x_1 = 1.2 - \frac{f(1.2)}{f'(1.2)}$$

$$= 1.2 - \frac{-0.073789\ldots}{1.563999\ldots} = 1.247180\ldots$$

Hence $x_1 = 1.247$ to 3 d.p.

Exercise 10C — Pages 284–285 Solution 95

4. $f(x) = 1 - x - \cos(x^2)$. Use radians.

a. Show that a root α lies in $1.4 < \alpha < 1.5$. (1 mark)

b. Starting with $x_0 = 1.4$, apply Newton–Raphson once, giving the second approximation to 3 d.p. (4 marks)

c. Use a sign change in a suitable interval to show that your answer to b is correct to 3 d.p. (2 marks)

4a

$f(1.4) = 1 - 1.4 - \cos(1.4^2) = -0.020548\ldots$

$f(1.5) = 1 - 1.5 - \cos(1.5^2) = 0.128173\ldots$

Continuity and the sign change give a root in $(1.4, 1.5)$.

4b

$f'(x) = -1 + 2x\sin(x^2)$.

$f'(1.4) = -1 + 2.8\sin 1.96 = 1.590592\ldots$

$$x_1 = 1.4 - \frac{f(1.4)}{f'(1.4)}$$

$$= 1.4 - \frac{-0.020548\ldots}{1.590592\ldots} = 1.412918\ldots$$

Hence $x_1 = 1.413$ to 3 d.p.

4c

$f(1.4125) = 1 - 1.4125 - \cos(1.4125^2)$

$= -0.000762437\ldots < 0$,

$f(1.4135) = 1 - 1.4135 - \cos(1.4135^2)$

$= 0.000811255\ldots > 0$.

Continuity and the sign change give a root between the rounding boundaries, so $\alpha = 1.413$ to 3 d.p.

Exercise 10C — Pages 284–285 Solution 96

5. $f(x) = x^2 - 3/x^2$, $x > 0$.

a. Show that a root α lies in $[1.3, 1.4]$. (1 mark)

b. Find $f'(x)$. (2 marks)

c. Starting with 1.3, apply Newton–Raphson once, giving the second approximation to 3 d.p. (3 marks)

5a

$$f(1.3) = 1.69 - \frac{3}{1.69} = -0.085147\ldots$$

$$f(1.4) = 1.96 - \frac{3}{1.96} = 0.429387\ldots$$

Continuity and the sign change give a root in $(1.3, 1.4)$.

5b

$$f'(x) = 2x + \frac{6}{x^3}.$$

5c

$$f'(1.3) = 2.6 + \frac{6}{1.3^3} = 5.330996\ldots$$

$$x_1 = 1.3 - \frac{f(1.3)}{f'(1.3)}$$

$$= 1.3 - \frac{-0.085147\ldots}{5.330996\ldots} = 1.315972\ldots$$

Hence $x_1 = 1.316$ to 3 d.p.

Exercise 10C — Pages 284–285 Solution 97

6. $y = f(x) = x^2 \sin x - 2x + 1$. P, Q and R are roots; A and B are stationary points with x-coordinates a, b.

a. Show a root lies in each interval: i. $[0.6, 0.7]$, ii. $[1.2, 1.3]$, iii. $[2.4, 2.5]$.

b. Explain why $x_0 = a$ is unsuitable for Newton–Raphson.

c. Starting with $x_0 = 2.4$, apply Newton–Raphson once. Give the second approximation to 3 d.p.

6a

$f(0.6) = 0.36 \sin 0.6 - 1.2 + 1 = 0.0032713\ldots$

$f(0.7) = 0.49 \sin 0.7 - 1.4 + 1 = -0.0843333\ldots.$

$f(1.2) = 1.44 \sin 1.2 - 2.4 + 1 = -0.0578637\ldots$

$f(1.3) = 1.69 \sin 1.3 - 2.6 + 1 = 0.0284133\ldots.$

$f(2.4) = 5.76 \sin 2.4 - 4.8 + 1 = 0.0906679\ldots$

$f(2.5) = 6.25 \sin 2.5 - 5 + 1 = -0.2595491\ldots.$

The function is continuous and changes sign on each interval, so each contains a root.

6b At the stationary point, $f'(a) = 0$. The formula would divide by zero, so $x_0 = a$ is unsuitable.

6c

$f'(x) = x^2 \cos x + 2x \sin x - 2.$

$f'(2.4) = 5.76 \cos 2.4 + 4.8 \sin 2.4 - 2$

$= -3.005165\ldots$

$$x_1 = 2.4 - \frac{f(2.4)}{f'(2.4)}$$

$$= 2.4 - \frac{0.090667\ldots}{-3.005165\ldots} = 2.430170\ldots$$

Hence $x_1 = 2.430$ to 3 d.p.

Exercise 10C — Pages 284–285 Solution 98

7. $f(x) = \ln(3x - 4) - x^2 + 10$, $x > 4/3$.

a. Show that a root α lies in $[3.4, 3.5]$.

b. Find $f'(x)$.

c. Starting with 3.4, apply Newton–Raphson once, giving the second approximation to 3 d.p.

7a

$f(3.4) = \ln 6.2 - 11.56 + 10 = 0.264549\ldots$

$f(3.5) = \ln 6.5 - 12.25 + 10 = -0.378197\ldots$

Continuity and the sign change give a root in $(3.4, 3.5)$.

7b

$$f'(x) = \frac{3}{3x - 4} - 2x.$$

7c

$$f'(3.4) = \frac{3}{6.2} - 6.8 = -6.316129\ldots$$

$$x_1 = 3.4 - \frac{f(3.4)}{f'(3.4)}$$

$$= 3.4 - \frac{0.264549\ldots}{-6.316129\ldots} = 3.441884\ldots$$

Hence $x_1 = 3.442$ to 3 d.p.

Application to Modelling

Solution 99

The price of a car, in pounds, x years after purchase is modelled by

$f(x) = 15000(0.85)^x - 1000\sin x, \quad x > 0.$

Use radians.

a. Find its value after 10 years, to the nearest £100.

b. Show that f has a root between 19 and 20.

c. Find $f'(x)$. Starting with 19.5, apply Newton–Raphson once to estimate when the value becomes zero. Give your answer to 3 d.p.

d. Criticise the model as the car gets older.

a

$f(10) = 3497.13\ldots$

The predicted value is £3500.

b

$f(19) = 534.114515\ldots > 0,$

$f(20) = -331.55\ldots < 0.$

Continuity and the sign change give a root in $(19, 20)$.

c

$f'(x) = 15000(0.85)^x \ln 0.85 - 1000\cos x.$

$f(19.5) = 25.0693\ldots, \quad f'(19.5) = -898.3009\ldots$

$x_1 = 19.5 - \dfrac{f(19.5)}{f'(19.5)} = 19.527907\ldots$

The second approximation is 19.528 years.

d

The model can predict negative prices, which are not meaningful car values. It is not reasonable for cars around 20 years old or older.

Exercise 10D — Page 287

Solution 100

1. An astronomer models a planet on an elliptical orbit. Its angle E radians is related to the angle M it would have travelled on a circular path by

   $M = E - 0.1\sin E, \quad E \geq 0.$

   She needs E when $M = \pi/6$.

a. Show that E is a root of $f(x) = x - 0.1\sin x - k$, finding k.

b. Starting with 0.6, apply Newton–Raphson once to find a second approximation to E.

c. Use a sign change in a suitable interval to show that the answer to b is correct to 3 d.p.

1a

$\dfrac{\pi}{6} = E - 0.1\sin E$

$E - 0.1\sin E - \dfrac{\pi}{6} = 0.$

Hence $k = \pi/6$.

1b

$f'(x) = 1 - 0.1\cos x.$

$x_1 = 0.6 - \dfrac{0.6 - 0.1\sin 0.6 - \pi/6}{1 - 0.1\cos 0.6}$

$= 0.6 - \dfrac{0.019936\ldots}{0.917466\ldots} = 0.578269\ldots.$

1c

$f(0.5775) = 0.5775 - 0.1\sin 0.5775 - \pi/6$

$= -0.000692\ldots < 0,$

$f(0.5785) = 0.5785 - 0.1\sin 0.5785 - \pi/6$

$= 0.000224361\ldots > 0.$

The function is continuous and changes sign between the rounding boundaries. Hence $E = 0.578$ to 3 d.p.